

## FES Stats Workshop

Overview of Stats process



## Outline for today

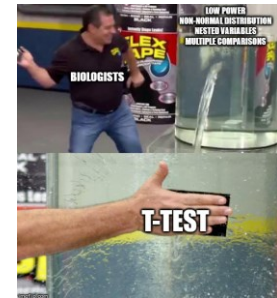
### Afternoon

Lecture [1.5 hours, including Q&A]

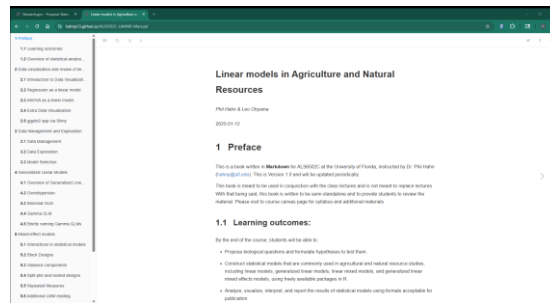
- Stats basics
- Overview of Stats process

Worked example in R and SAS

- [1.5 hours, including Q&A]

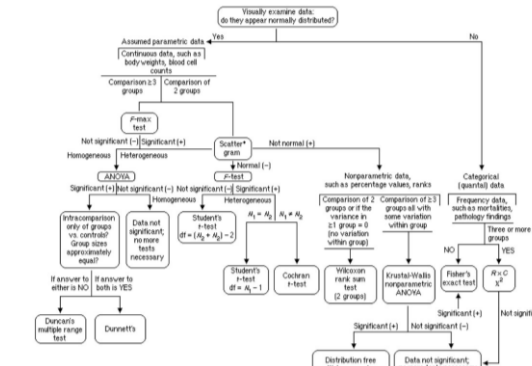


## ALS6502C Course Manual



<https://hahnp13.github.io/ALS6502C-LMANR-Manual/>

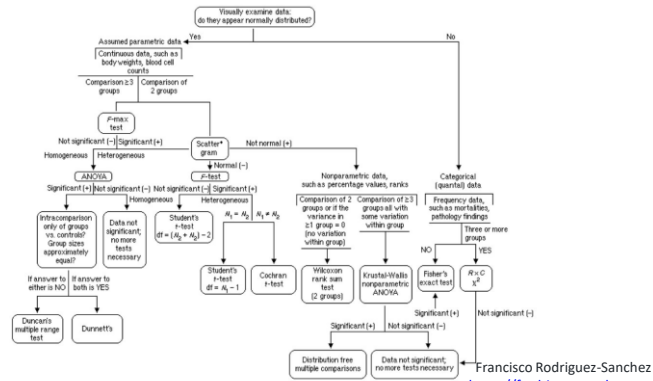
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Francisco Rodriguez-Sanchez  
<https://frodriquezsanchez.net>

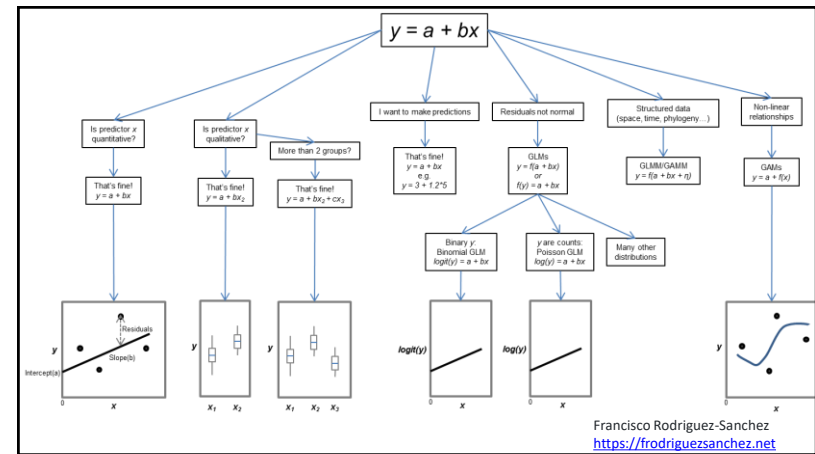
4

## Modern statistics are easier than this



Francisco Rodriguez-Sanchez  
<https://frodriguezsanchez.net>

5



Francisco Rodriguez-Sanchez  
<https://frodriguezsanchez.net>

6



7

### What is a probability distribution?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

8

### What is a mean?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

10

### What is a standard deviation?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

12

### What is a standard error?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

14

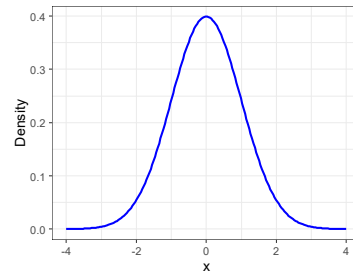
### Questions ???

16

## What is a probability distribution?

### Normal (Gaussian) distribution

- Continuous numeric variables
  - $-\infty$  to  $\infty$
  - Most points near the mean
  - Points symmetrically distributed around mean

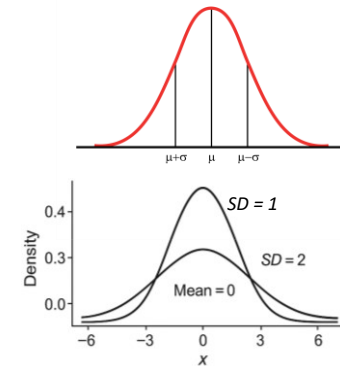


17

## What is a probability distribution?

### Normal (Gaussian) distribution

- Continuous numeric variables
  - $-\infty$  to  $\infty$
  - Most points near the mean
  - Points symmetrically distributed around mean
- Parameters
  - $\mu$  = Mean
  - $\sigma^2$  = Variance
    - $\sigma$  = Standard deviation



18

## What processes generate data?

- Data are the result of natural (or biological) phenomena

Counts



Proportion



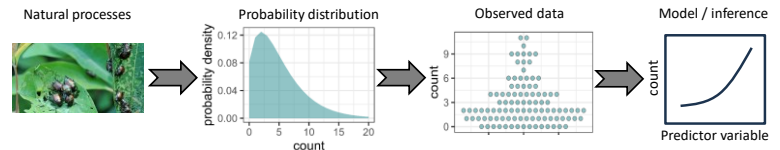
Time-to-Event



19

## What processes generate data?

- Data are the result of natural (or biological) phenomena
- Phenomena is mathematically described by a probability distribution
- Data we observe are a random subset of events produced by a phenomenon
- Used to model data in statistical analysis and make inferences



20



## Steps for constructing and interpreting a GLMM

0. Data Exploration
1. Construct model
  - a. Fixed effects (predictor or explanatory variables)
  - b. Random effects (blocking variables)
  - c. Error distribution (response variable or dependent variable)
2. Assess residuals and overdispersion
3. Check random effects and/or distribution using AIC (not always needed)
4. Examine significance of terms in model
  1. Anova
  2. AIC model selection
5. Calculate treatment means and/or slopes
6. Construct contrasts
7. Calculate R<sup>2</sup><sub>m</sub> and R<sup>2</sup><sub>c</sub>
8. Make pretty figure

21

## Methods in Ecology and Evolution



*Methods in Ecology and Evolution* 2016, 7, 636–645

doi: 10.1111/2041-210X.12577

SPECIAL FEATURE: 5<sup>TH</sup> ANNIVERSARY OF *METHODS IN ECOLOGY AND EVOLUTION*

### A protocol for conducting and presenting results of regression-type analyses

Alain F. Zuur<sup>1\*</sup> and Elena N. Ieno<sup>2</sup>

<sup>1</sup>Highland Statistics Ltd., 9 St Clair Wynd, Newburgh AB41 6DZ, UK; and <sup>2</sup>Highland Statistics Ltd., Av Finlandia 11, Bwg 34, Santa Pola, Spain



22

## Step 0: Data exploration

Plot data, check for outliers/errors.

Check distribution, etc.

- Justify subsetting, transformation decisions *a priori*
  - Or at least for biological reasons
- Avoid making decisions based on your specific dataset

Methods in Ecology and Evolution 2010, 1, 3–14

doi: 10.1111/j.2041-210X.2009.00001.x

### A protocol for data exploration to avoid common statistical problems

Alain F. Zuur<sup>1,2</sup>, Elena N. Ieno<sup>3,4</sup> and Chris S. Elphick<sup>3</sup>

<sup>1</sup>Highland Statistics Ltd, Newburgh, UK; <sup>2</sup>Oceanlab, University of Aberdeen, Newburgh, UK; and <sup>3</sup>Department of Ecology and Evolutionary Biology and Center for Conservation Biology, University of Connecticut, Storrs, CT, USA



|                         |                                                                        |                                                    |
|-------------------------|------------------------------------------------------------------------|----------------------------------------------------|
| 1                       | Formulate biological hypothesis<br>Carry out experiment & collect data |                                                    |
| <b>Data exploration</b> |                                                                        |                                                    |
| 1.                      | Outliers Y & X                                                         | boxplot & Cleveland dotplot                        |
| 2.                      | Homogeneity Y                                                          | conditional boxplot                                |
| 3.                      | Normality Y                                                            | histogram or QQ-plot                               |
| 2                       | 4. Zero trouble Y                                                      | frequency plot or correlogram                      |
|                         | 5. Collinearity X                                                      | VIF & scatterplots<br>correlations & PCA           |
|                         | 6. Relationships Y & X                                                 | (multi-panel) scatterplots<br>conditional boxplots |
|                         | 7. Interactions                                                        | coplots                                            |
|                         | 8. Independence Y                                                      | ACF & variogram<br>plot Y versus time/space        |
| 3                       | Apply statistical model                                                |                                                    |

23

## Step 1. Construct a statistical model to test your hypothesis

1. Construct a statistical model to test your hypothesis.

- Linear model:  $y \sim x_1 + x_2 + x_1:x_2$
- Random effects?
- Distribution? Gaussian okay or generalized model?

24

### Step 1. Construct a statistical model to test your hypothesis

- Linear model:

$$y_i \sim \mu + \beta + e_i$$

Response variable      mean      Residual variation or error

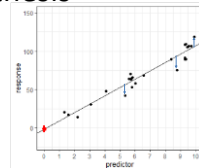
Parameter estimate to change the mean based on some variable (ie. slopes or intercepts)

25

### Step 1. Construct a statistical model to test your hypothesis

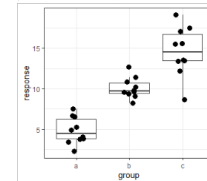
- “Linear regression”:  $y_i \sim B_0 + B_1 X + e_i$

intercept      slope      resid



- “ANOVA” model:  $y_{ij} \sim B_0 + \beta_i + e_{ij}$

intercept      resid  
Adjustment to intercept for each group



26

Questions ???



## Linear mixed models

- “ANOVA” model:

$$y_{ij} \sim B_0 + \beta_i + e_{ij}$$

- “RCBD”:

$$y_{ij} \sim B_0 + \beta_i + \alpha_j + e_{ij}$$

response

Predictor variable or fixed effects  
(ie. different intercepts for  
treatment means)

Random effect (ie. random intercepts for blocks)

## Fixed vs. random effects?

Fixed effects

Random effects

32

## Fixed vs. random effects?

**Fixed effects**

**Random effects**

- Usually the factor(s) of interest in study (or covariates)
- Same anywhere you do it
  - Values are meaningful and not exchangeable
- Represent all possible levels of a factor (of interest)
  - Same for continuous variables
- **Conclusions restricted to those levels (group effects)**

33

## Fixed vs. random effects?

### Fixed effects

- Usually the factor(s) of interest in study (or covariates)
- Same anywhere you do it
  - Values are meaningful and not exchangeable
- Represent all possible levels of a factor (of interest)
  - Same for continuous variables
- **Conclusions restricted to those levels (group effects)**

### Random effects

- Usually not the focus of study
- Levels (categories) could be exchanged/substituted (eg. Individuals or blocks)
- Represent a (random) sample of a larger set of levels
- **Can extrapolate conclusions to the larger sample (population level)**
- Different statistical properties

## Benefits of blocking

- Blocks are formed so that treatments within blocks are as homogenous as possible
- Treatments randomly assigned within blocks
- Used to reduce variation among experimental units
  - Allows to account for variation from block-to-block
- Ensures interspersed among treatments
- Can account for pseudo-replication (when present)

### Rules of thumb about random effects

- Sometimes it's a judgement call of whether something should be fixed or random
- Random effects should have at least 5 levels (and typically much more)
  - 5+ blocks, 5+ populations, 5+ clutches
- Each level of a random effect should have >3 observations
  - For most levels
  - Ok if a few levels have only a few observations
  - *Should be relatively balanced among levels* (but not perfectly)

Harrison et al. (2018), A brief introduction to mixed effects modelling and multi-model inference in ecology. PeerJ 6:e4794; DOI 10.7717/peerj.4794

### Examples of blocks

- Chambers or tanks



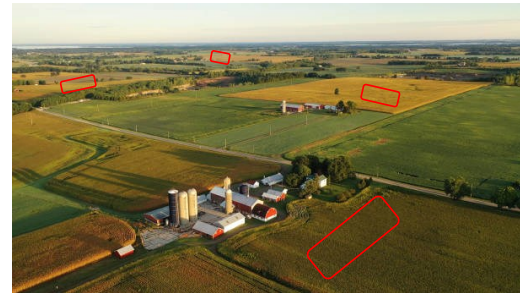
### Examples of blocks

- Fields



### Examples of blocks

- Farms



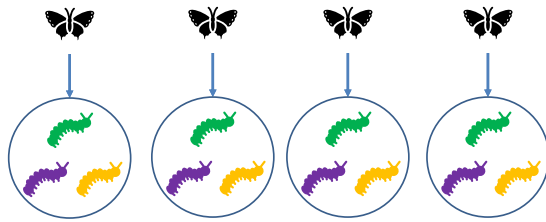
39

40



### Examples of blocks

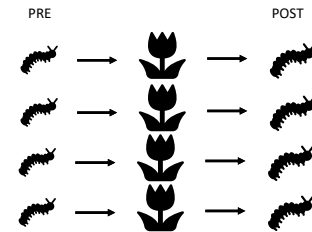
- Parent – siblings all similar to each other



41

### Examples of blocks

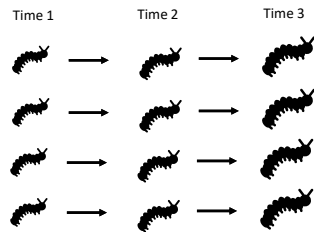
- Individuals/subjects
- Repeated measures



42

### Examples of blocks

- Individuals/subjects
- Repeated measures



43



44

Which of the following would be characteristics typical of a **FIXED** effect in a statistical model? (select all that apply)

1. A factor of direct interest hypothesized to affect your response variable
2. A factor not of direct interest to the study which might have an unintended effect on your response variable
3. Values or levels are meaningful and the same anywhere you conducted the experiment
4. Values or levels could be exchanged or substituted
5. Values or levels represent a random subset of possible levels of the factor
6. Conclusions are restricted to the levels imposed
7. Conclusions can be extrapolated to the larger population

45

Which of the following would be characteristics typical of a **FIXED** effect in a statistical model? (select all that apply)

- 1. A factor of direct interest hypothesized to affect your response variable**
2. A factor not of direct interest to the study which might have an unintended effect on your response variable
- 3. Values or levels are meaningful and the same anywhere you conducted the experiment**
4. Values or levels could be exchanged or substituted
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- 6. Conclusions are restricted to the levels imposed**
7. Conclusions can be extrapolated to the larger population

46

Which of the following would be characteristics typical of a **RANDOM** effect in a statistical model? (select all that apply)

1. A factor of direct interest hypothesized to affect your response variable
2. A factor not of direct interest to the study which might have an unintended effect on your response variable
3. Values or levels are meaningful and the same anywhere you conducted the experiment
4. Values or levels could be exchanged or substituted
5. Values or levels represent a random subset of possible levels of the factor
6. Conclusions are restricted to the levels imposed
7. Conclusions can be extrapolated to the larger population

47

Which of the following would be characteristics typical of a **RANDOM** effect in a statistical model? (select all that apply)

1. A factor of direct interest hypothesized to affect your response variable
2. **A factor not of direct interest to the study which might have an unintended effect on your response variable**
3. Values or levels are meaningful and the same anywhere you conducted the experiment
4. **Values or levels could be exchanged or substituted**
5. **Values or levels represent a random subset of possible levels of the factor**
6. Conclusions are restricted to the levels imposed
7. **Conclusions can be extrapolated to the larger population**

48

## Step 1. Construct a statistical model to test your hypothesis

Table of R functions for constructing linear models

| R function | SAS proc                           | Type of model                                                          | Type of test                                                     | Distribution                              |
|------------|------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------|
| lm()       | Proc glm<br>Proc anova<br>Proc reg | Linear model                                                           | Regression, ANOVA, ANCOVA                                        | Gaussian                                  |
| lmer()     | Proc mixed                         | Linear mixed-effects models                                            | Blocked or split-plot ANOVAs, nested designs, repeated measures. | Gaussian                                  |
| glmer()    | Proc glimmix                       | Generalized linear mixed-effects models                                | Blocked or split-plot ANOVAs, nested designs, repeated measures. | Poisson, Binomial                         |
| glmmTMB()  | Proc glimmix                       | Any type of linear models, including generalized linear (mixed) models | Blocked or split-plot ANOVAs, nested designs, repeated measures. | Poisson, Binomial, Gamma, and many others |

49

Break



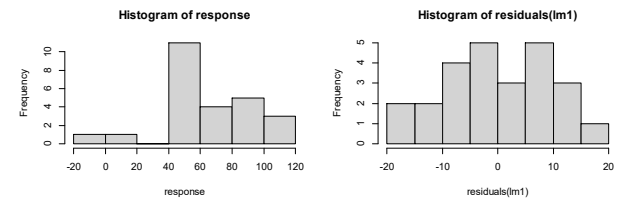
50

## Step 2. Check model assumptions

51

## Assumptions of linear models

- **Response variable errors (ie. residuals) normally distributed**

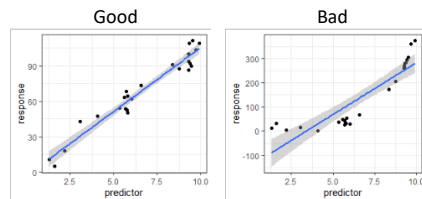


Residuals should be normally distributed

52

## Assumptions of linear models

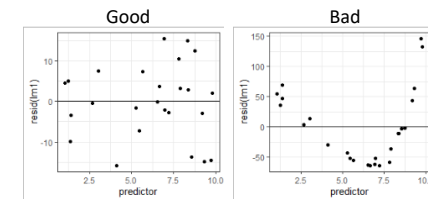
- Response variable errors (ie. residuals) normally distributed
- **Linearity**



53

## Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- **Homoscedasticity**

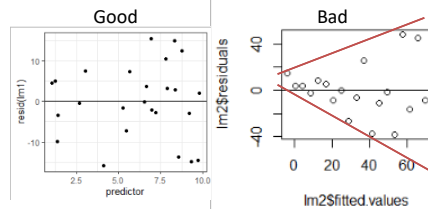


Residuals should be evenly scattered around zero when plotted against the original predictor

54

## Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- **Homoscedasticity**

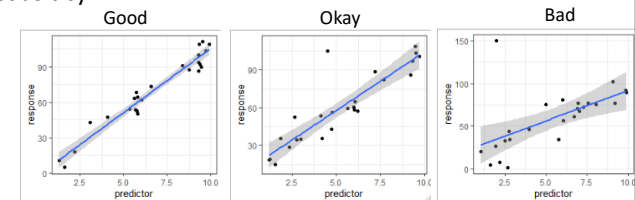


Residuals should be evenly scattered around zero when plotted against the original predictor

55

## Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- Homoscedasticity
- **Outliers**



56



Often data (and residuals)  
are not normally distributed

57

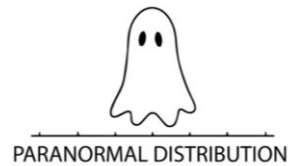
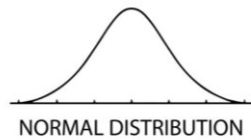
Why might we need to use non-normal distributions?

- Linear models are *reasonably* robust to violations of the assumptions
- Violations can affect
  - P-values
  - Standard errors
  - Parameter estimates
- Generalized models provide a better fit to the data
  - P-values
  - Standard errors
  - Parameter estimates
- Reviewers and editors expect it



58

Questions ???



59

What are “generalized” models?

- Linear models that allow for more flexible error distributions
- Parameters estimated via maximum likelihood (not least squares)
- Model components
  - Linear predictor formula (  $y \sim x$  )
  - Error distribution (usually exponential family)
  - Link function: links distribution to linear predictor formula
    - ie. Transforms error distribution to linear
- Generalized linear **mixed** model
  - Allows fixed and random effects

60

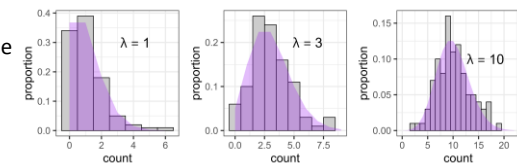
## Generalized linear mixed models

- Linear model:  $y_{ij} \sim \mu + \beta_i + \varepsilon_{ij}$
- Linear mixed model:  $y_{ij} \sim \mu + \alpha_i + \beta_j + \varepsilon_{ij}$
- Generalized linear mixed model:  $\log(y_{ij}) \sim \mu + \alpha_i + \beta_j + \varepsilon_{ij}$

61

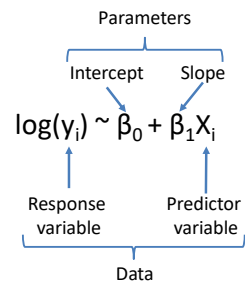
## Poisson distribution - assumptions

- Discrete probability distribution
  - Integers: number of events (per unit of time or space; ie. rate)
  - Events occur randomly
  - Events are independent
- Parameters
  - $\lambda$  = mean = variance
  - Link function = log

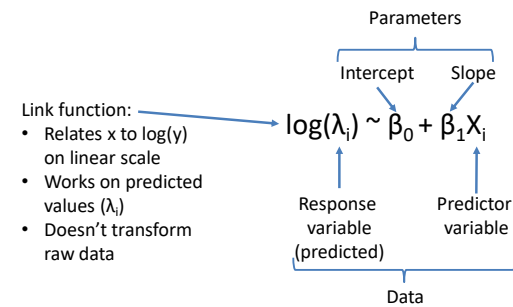


62

### Poisson linear model



### Poisson linear model



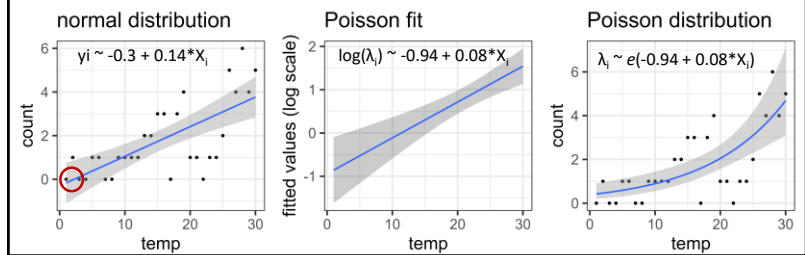
## Poisson linear model

$$\begin{aligned}\log(\lambda_i) &\sim \beta_0 + \beta_1 X_i \\ &= \\ \lambda_i &\sim e^{\beta_0 + \beta_1 X_i}\end{aligned}$$

65

## Poisson linear model – example 1

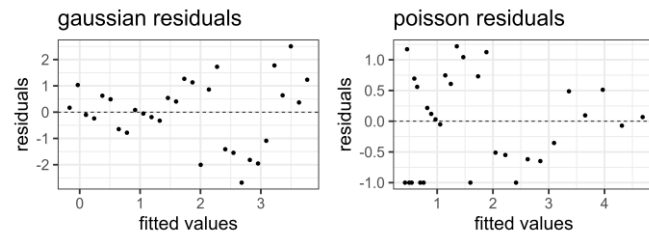
- Counts across temp (continuous, 1 to 30):
- $\log(\lambda_i) \sim \beta_0 + \beta_1 X_i$



66

## Poisson linear model – example 1

- Counts across temp (continuous, 1 to 30):
- $\log(\lambda_i) \sim \beta_0 + \beta_1 X_i$

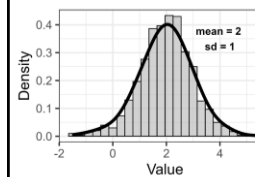


67

## Common distributions

### Normal (gaussian)

- Continuous numeric
- Parameters
  - Mean
  - Variance (SD)
- Link = identity (none)

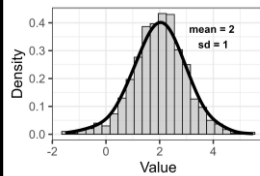


68

## Common distributions

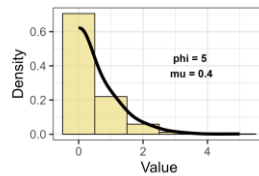
### Normal (gaussian)

- Continuous numeric
- Parameters
  - Mean
  - Variance (SD)
- Link = identity (none)



### Negative binomial

- Counts
- Parameters
  - $\mu$  (mean)
  - $\phi$  (precision)
    - Inverse of dispersion
- Link = log

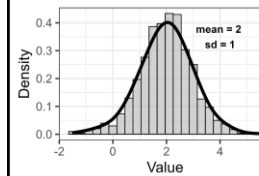


69

## Common distributions

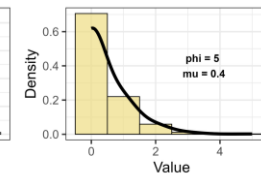
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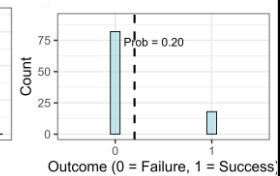
### Negative binomial

- Counts
- Parameters
  - $\mu$  (mean)
  - $\phi$  (precision)
    - Inverse of dispersion
- Link = log



### Binomial

- Proportions
  - **Binary** or ratio
- Parameters
  - Probability
  - # trials
- Link = logit

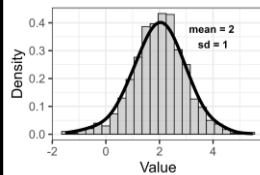


70

## Common distributions

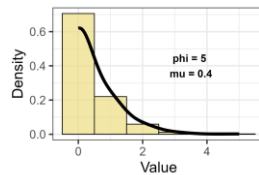
### Normal (gaussian)

- Continuous numeric
- Parameters
  - Mean
  - Variance (SD)
- Link = identity (none)



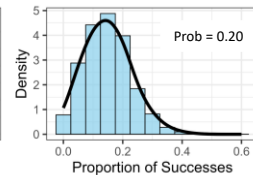
### Negative binomial

- Counts
- Parameters
  - $\mu$  (mean)
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- Link = log



### Binomial

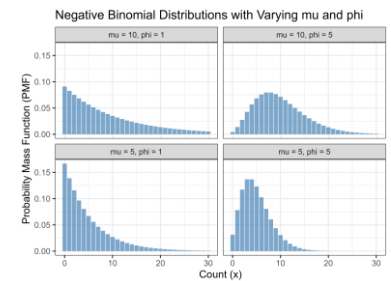
- Proportions
  - Binary or ratio
- Parameters
  - Probability
  - # trials
- Link = logit



71

## Negative binomial distribution - assumptions

- Discrete probability distribution
  - Integers: number of events
  - Events occur randomly (follow Poisson distribution)
  - Account for “extra” variation (beyond poisson)
- Parameters
  - $\mu$  = mean
  - $\phi$  = overdispersion parameter
    - lower values are more overdispersion
    - higher values converge to Poisson
  - $\text{Var} = \mu + \mu^2 / \phi$
  - Link function = log



72



## Common types of distributions

| Distribution      | Type of data                                                    | Overdispersion parameter? | Link           |
|-------------------|-----------------------------------------------------------------|---------------------------|----------------|
| Gaussian          | Continuous numeric variable                                     | Yes                       | Identity       |
| Log-normal        | Continuous numeric variable                                     | Yes                       | Identity       |
| Poisson           | Integers, counts                                                | No                        | Natural log    |
| Negative binomial | Integers, counts                                                | Yes                       | Natural log    |
| Binomial          | Success/failure or presence/absence (binary), ratio             | No                        | Logit, cloglog |
| Beta-binomial     | Success/failure, presence/absence (binary), ratio               | Yes                       | Logit, cloglog |
| Beta              | Proportion (when the denominator is not known)                  | Yes                       | Logit          |
| ordbeta           | Proportion (when the denominator is not known), can have zero's | Yes                       | Logit          |
| Gamma             | Positive continuous variable                                    | Yes                       | Inverse, log   |
| ziGamma           | Positive continuous variable, can have zero's                   | Yes                       | log            |

73

## What distribution best matches the data?

### Description of data

1. Concentrations of chemicals in the soil, most values are low ( $\sim 0.1$  ug/L) but some are high (5-10 ug/L)
2. Number of infected and uninfected aphids on plants
3. Percent damage on leaves caused by herbivore or pathogens
4. Number of thrips on a plants

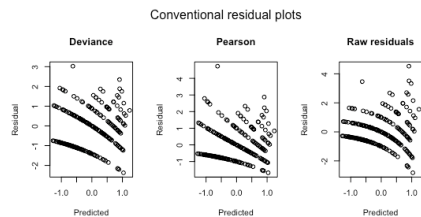
### Distribution

- A. Normal
- B. Log-normal
- C. Poisson
- D. Negative binomial
- E. Binomial
- F. Beta
- G. Gamma

74

## Step 2. Check GLM assumptions

- “Conventional” residuals don’t work well for non-normal data



This figure shows residuals fit for count data assuming a Poisson distribution. The model (not shown) fits the data well, although the residuals look wonky.

75

## Step 2. Check GLM assumptions

- “Regular” residuals often don’t work well for non-normal data
- Simulated residuals offer a great diagnostic tool

### DHARMA: residual diagnostics for hierarchical (multi-level/mixed) regression models

Florian Hartig, Theoretical Ecology, University of Regensburg [website](#)  
2021-03-27

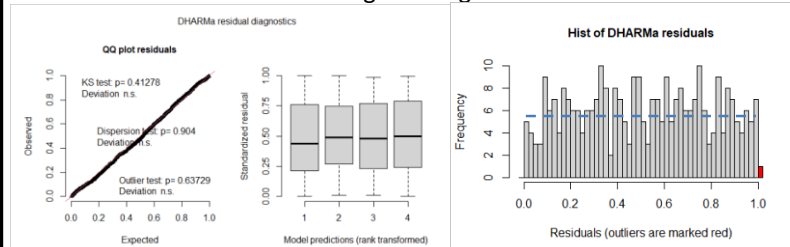
DHARMA stands for “Diagnostics for Hierarchical Regression Models”

<https://cran.r-project.org/web/packages/DHARMA/vignettes/DHARMA.html>

76

## Step 2. Check GLM assumptions

- “Regular” residuals often don’t work well for non-normal data
- Simulated residuals offer a great diagnostic tool



<https://cran.r-project.org/web/packages/DHARMa/vignettes/DHARMa.html>

77

## Reasons for deviations of residuals

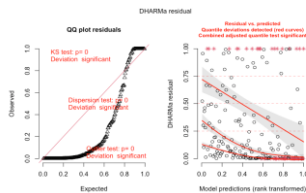
- Mis-specified model
  - Incorrect distribution (or potentially link function)
  - Missing fixed effect
    - Include additional covariates, if available
    - Check outliers in covariates or influential data points
    - Include quadratic term
  - Incorrect random-effect structure
    - Missing block or nestedness?
    - Incorrect order of nesting?
- **Overdispersion** (return to this later)
- Zero-inflation

78

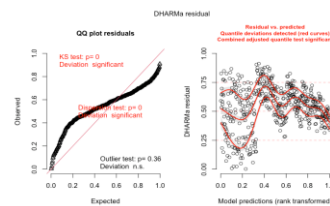
## Step-by-step guide to analysis (post-data collection)

### 2. Check model assumptions

Bad residuals (overdispersion)



Bad residuals (wrong dist., overfitting, etc.)



79

## Step 3. Check random effects and model components

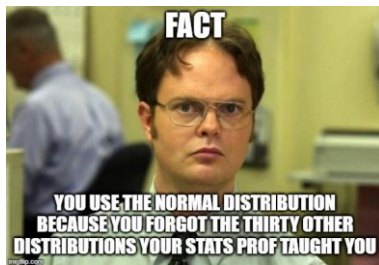
### 3. Check random effects using AIC or LRT

- Not always necessary
  - Not all models have/need random effects and non-normal distributions
- Can use `summary()` to look at random effect variance components
  - Singularity can be worth noting (very small VC is *almost* singular)

**Will demo this in worked example**

84

Questions ???



85

### Step 4. Check significance of terms in model

Where do p-values come from?

- Estimate parameter values (i.e., slopes or intercepts)
- Calculate a test-statistic
- Use a distribution to find your p-value

Alternative method is AIC model selection

86

## Methods to estimate parameters

### Least squares

- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$

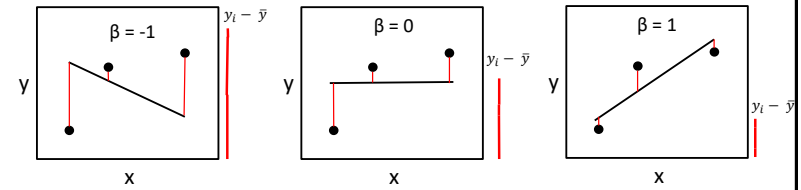
### Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$

88

## Least squares estimation

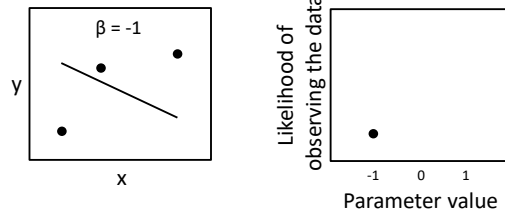
- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$
- $\beta$  = slope; Fit represents  $\bar{y}$



89

### Maximum likelihood estimation

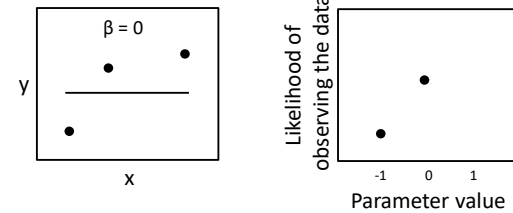
- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$ : How likely is  $\beta$  given the observed data ( $y$ )
- $\beta$  = slope



90

### Maximum likelihood estimation

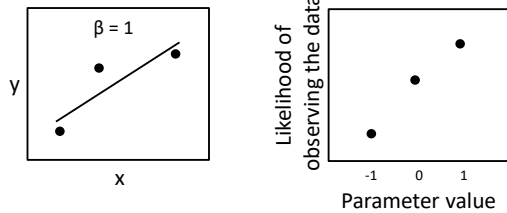
- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$ : How likely (probable) is  $\beta$  given the observed data ( $y$ )
- $\beta$  = slope



91

## Maximum likelihood estimation

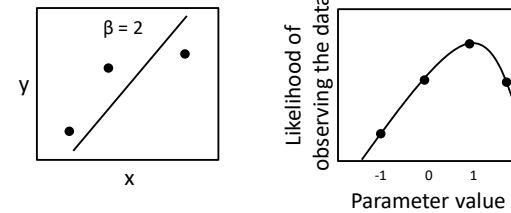
- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$ : How likely (probable) is  $\beta$  given the observed data ( $y$ )
- $\beta$  = slope



92

## Maximum likelihood estimation

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$ : How likely (probable) is  $\beta$  given the observed data ( $y$ )
- $\beta$  = slope



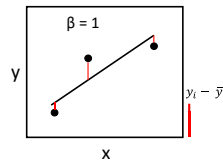
93



## Methods to estimate parameters

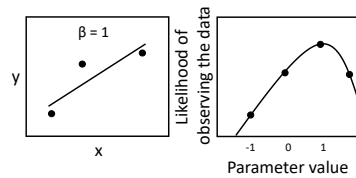
### Least squares

- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$
- $\beta$  = slope or intercept adjust
- Only works for normal data



### Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$
- $\beta$  = slope or intercept adjust
- Can be generalized to any distribution

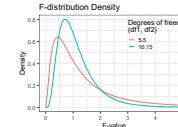


94

## Methods to estimate parameters

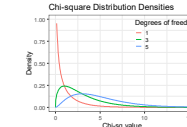
### Least squares

- Minimizes the sums of squares
- $\sum (y_i - \bar{y})^2$
- $\beta$  = slope or intercept adjust
- Only works for normal data
- F-value for significance test
  - $F = MS_B / MS_W$



### Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$
- $\beta$  = slope or intercept adjust
- Can be generalized to any distribution**
- $\chi^2$  for significance test
  - $\chi^2 = (\text{effect size} / SE)^2$



95

### After break

- Worked example
- Overview Steps 1-4
- **Step 5: Calculate group means**
- **Step 6: Conduct contrasts**
- **Step 7: Calculate R<sup>2</sup>**
- Step 8: Visualize data (if time permits)

100



109